MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE EBE

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital multiplexer module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules. Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.0 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/173-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

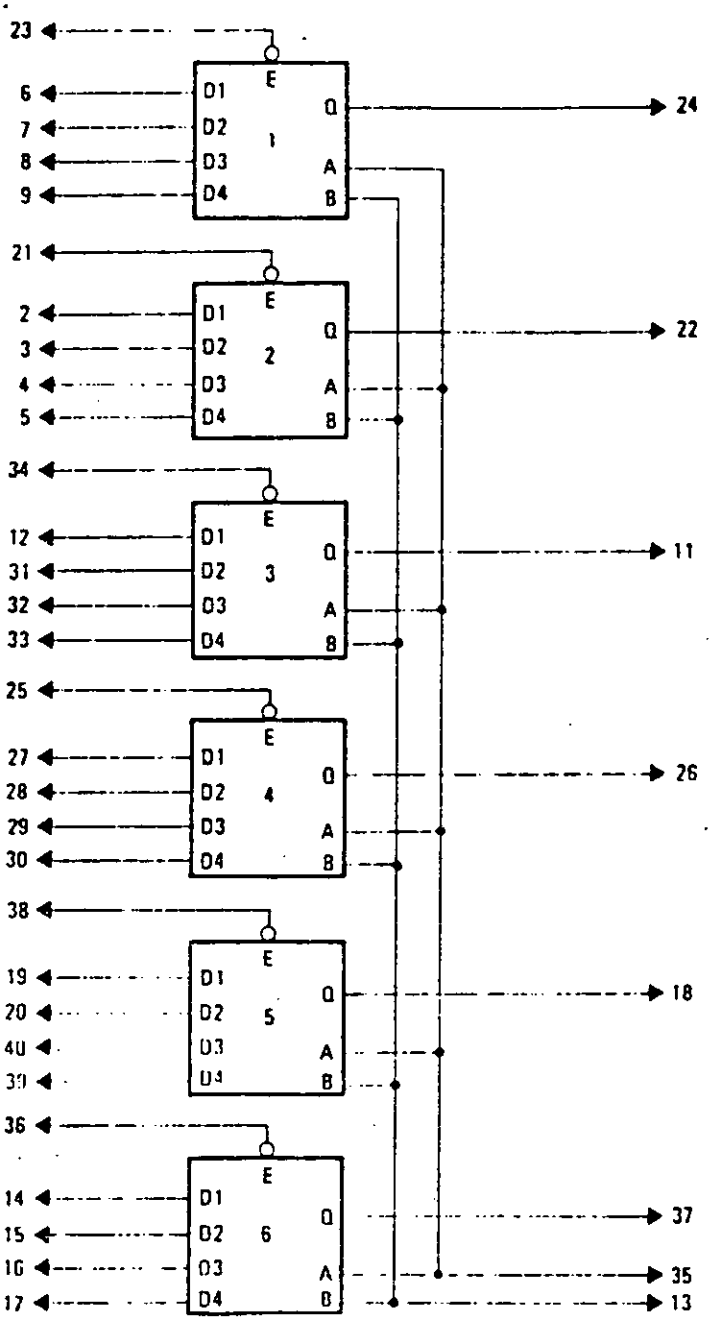
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
D1 - D4 = Data inputs.
E = Enable inputs.
A, B = Data select inputs.
Q = Output.

FIGURE 1. EBE module function diagram, digital multiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	V_{OUT} (low-loaded)	III	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
B1,B2	Input breakdown current, select	IV	S	0	200	0	200	0	200	0	200	µA
B3-B32	Input breakdown current, enable and data	IV	S	0	1	0	1	0	1	0	1	mA
C1-C30	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation, frame	VI	S	10	∞	1	∞	10	∞	1	∞	MΩ
E1-E30	Static output voltage, low level	VII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
F1-F24	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1,G2	Input current, select, low level	VIII	X	0.05	0.40	0.05	0.40	0.05	0.40	0.05	0.40	mA
G3-G32	Input current, data and enable, low level	VIII	X	0.1	0.9	0.1	0.9	0.1	0.9	0.1	0.9	mA
H1,H2	Input current, select, high level	IX	X	0	20	0	20	0	20	0	20	µA
H3-H32	Input current, data and enable, high level	IX	X	0	20	0	20	0	20	0	20	µA
J1-J6	Output short circuit current	X	X	10	28	10	28	10	28	10	28	mA
K	Power supply current	XI	S	5	81	5	81	5	81	5	81	mA
L1-L6	Transition time (t_{TLH}), C_L = 40 pF	XII	S	2	25	2	25	2	25	2	25	ns
L7-L12	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	5	45	5	45	5	45	5	45	ns
M1-M6	Transition time (t_{THL}), C_L = 40 pF	XIII	S	1	20	1	20	1	20	1	20	ns
M7-M12	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	2	30	2	30	2	30	2	30	ns
N1-N6	Propagation delay time (t_{PLH}), data to output, C_L = 40 pF	XIV	S	5	40	5	40	5	50	5	50	ns
N7-N24	Propagation delay time (t_{PLH}), data to output, C_L = 40 pF	XIV	Q	5	40	5	40	5	50	5	50	ns
N25-N36	Propagation delay time (t_{PLH}), select to output, C_L = 40 pF	XIV	S	5	120	5	120	5	140	5	140	ns
N37-N42	Propagation delay time (t_{PLH}), enable to output, C_L = 40 pF	XIV	S	5	66	5	66	5	75	5	75	ns

See footnotes at end of table.

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TABLE 1. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ - 3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N43-N48	Propagation delay time (t_{PLH}), data to output, C_L = 140 pF	XIV	Q	5	60	5	60	5	70	5	70	ns
N49-N60	Propagation delay time (t_{PLH}), select to output, C_L = 140 pF	XIV	Q	5	140	5	140	5	160	5	160	ns
N61-N66	Propagation delay time (t_{PLH}), enable to output, C_L = 140 pF	XIV	Q	5	86	5	86	5	95	5	95	ns
P1-P6	Propagation delay time (t_{PHL}), data to output, C_L = 40 pF	XV	S	5	50	5	50	5	55	5	55	ns
P7-P24	Propagation delay time (t_{PHL}), data to output, C_L = 40 pF	XV	Q	5	50	5	50	5	55	5	55	ns
P25-P36	Propagation delay time (t_{PHL}), select to output, C_L = 40 pF	XV	S	5	120	5	120	5	140	5	140	ns
P37-P42	Propagation delay time (t_{PHL}), enable to output, C_L = 40 pF	XV	S	5	50	5	50	5	55	5	55	ns
P43-P48	Propagation delay time (t_{PHL}), data to output, C_L = 140 pF	XV	Q	5	70	5	70	5	75	5	75	ns
P49-P60	Propagation delay time (t_{PHL}), select to output, C_L = 140 pF	XV	Q	5	140	5	140	5	160	5	160	ns
P61-P66	Propagation delay time (t_{PHL}), enable to output, C_L = 140 pF	XV	Q	5	70	5	70	5	75	5	75	ns
R1-R6	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R7-R12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVII	Q	0	300	0	300	0	300	0	300	mV
T	Power supply filter capacitance	XVIII	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U30	Input capacitance, data and enable	XIX	Q	2	15	2	15	2	15	2	15	pF
U31,U32	Input capacitance, select	XIX	Q	2	25	2	25	2	25	2	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B or D are part of a sequence of tests, the order of testing shall be as follows: A, B, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C and E through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C and E through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C and E through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/173-1 supersedes part number M28787/173. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	444 mW (outputs not loaded)
Supply voltage	5 Vdc ±10%
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Data and enable input current low level	
(V_{IL} = 0.0 V)	0.9 mA maximum
(V_{IL} = 0.4 V)	0.8 mA maximum

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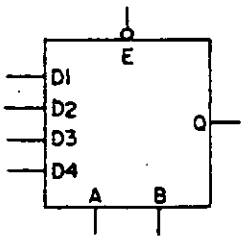
Data and enable input current high level	
(V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Select input current low level	
(V_{IL} = 0.0 V)	0.4 mA maximum
(V_{IL} = 0.3 V)	0.36 mA maximum
Select input current high level	
(V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	200 µA maximum
Output voltage	
Low level (I_{OL} = 8.0 mA)	0.4 V maximum
High level (I_{OH} = -400 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	8.0 mA
High level (V_{OH} = 2.4 V)	-400 µA

6.5 Module function. This module contains six 4 to 1 digital multiplexers as illustrated in figure 1. Each multiplexer has 4 input data lines (D1 to D4) plus an enable input (E) and uses two common data select lines (A and B). The data line used is determined by the state of the data select inputs. The data is transferred to the output (Q) providing the enable input is low. When the enable is high, the output is low regardless of all other inputs.

6.5.1 Input to output relationship. Each low-power multiplexer has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

SELECT A	SELECT B	ENABLE E	OUTPUT Q
L	L	L	D4
L	H	L	D3
H	L	L	D2
H	H	L	D1
L	L	H	L
L	H	H	L
H	L	H	L
H	H	H	L



1/ L = Low level.
2/ H = High level.
3/ D1-D4 = Data inputs.

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible.

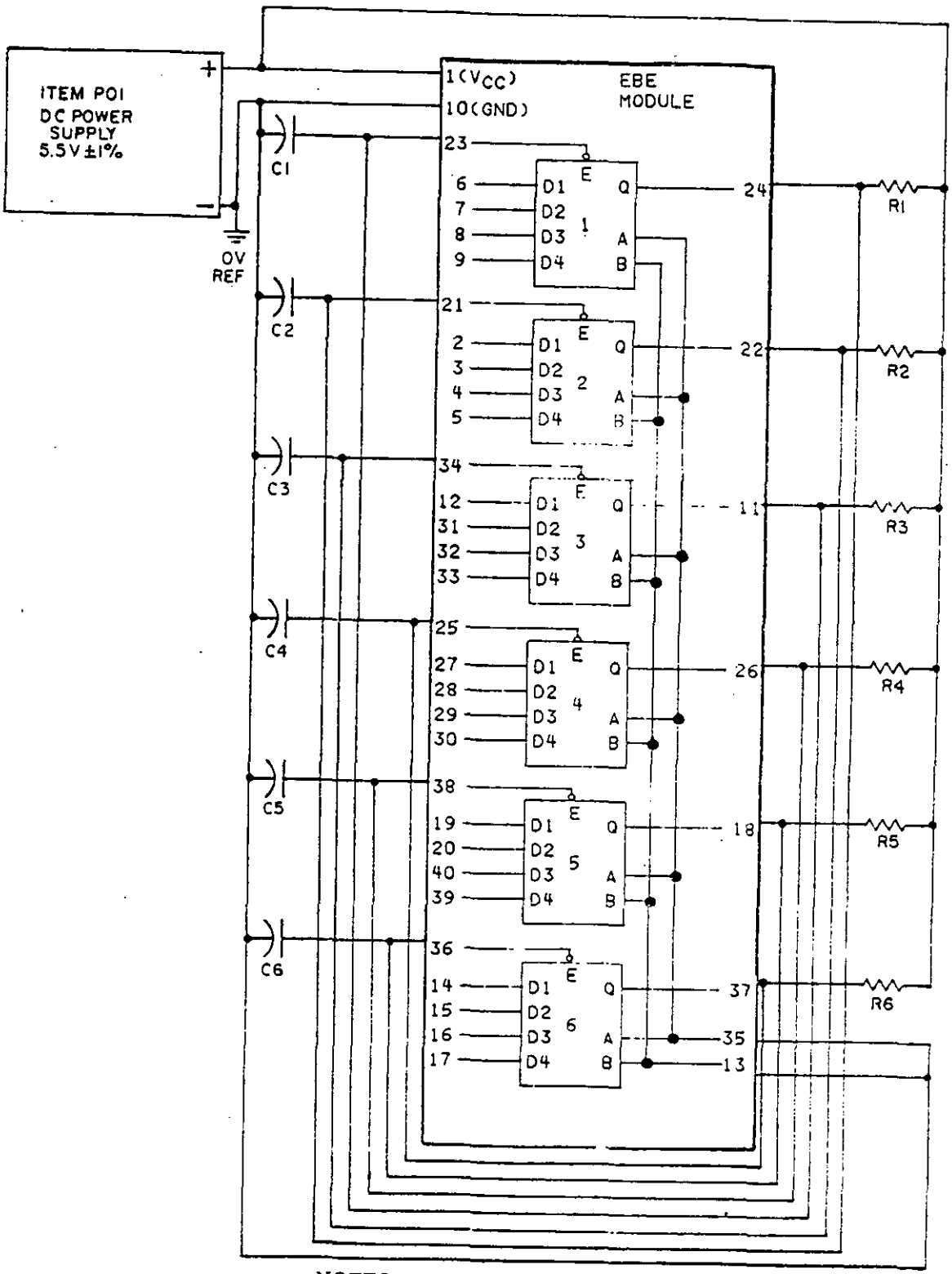
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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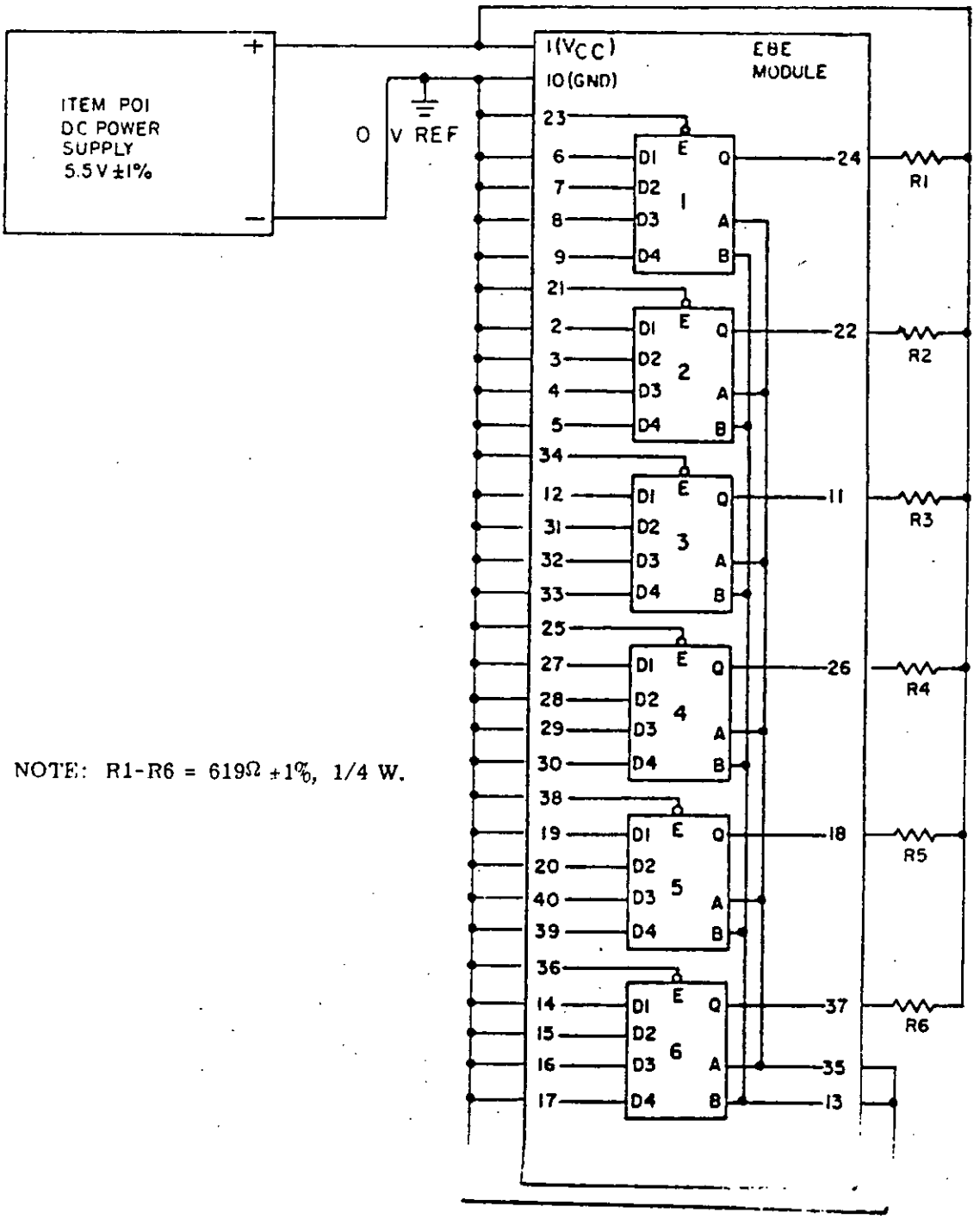
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NOTES:
1. R1-R6 = 619Ω ±1%, 1/4 W.
2. C1-C6 = 100 pF ±20%.

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FIGURE 2. Operating temperature and life test circuit.



NOTE: R1-R6 = 619Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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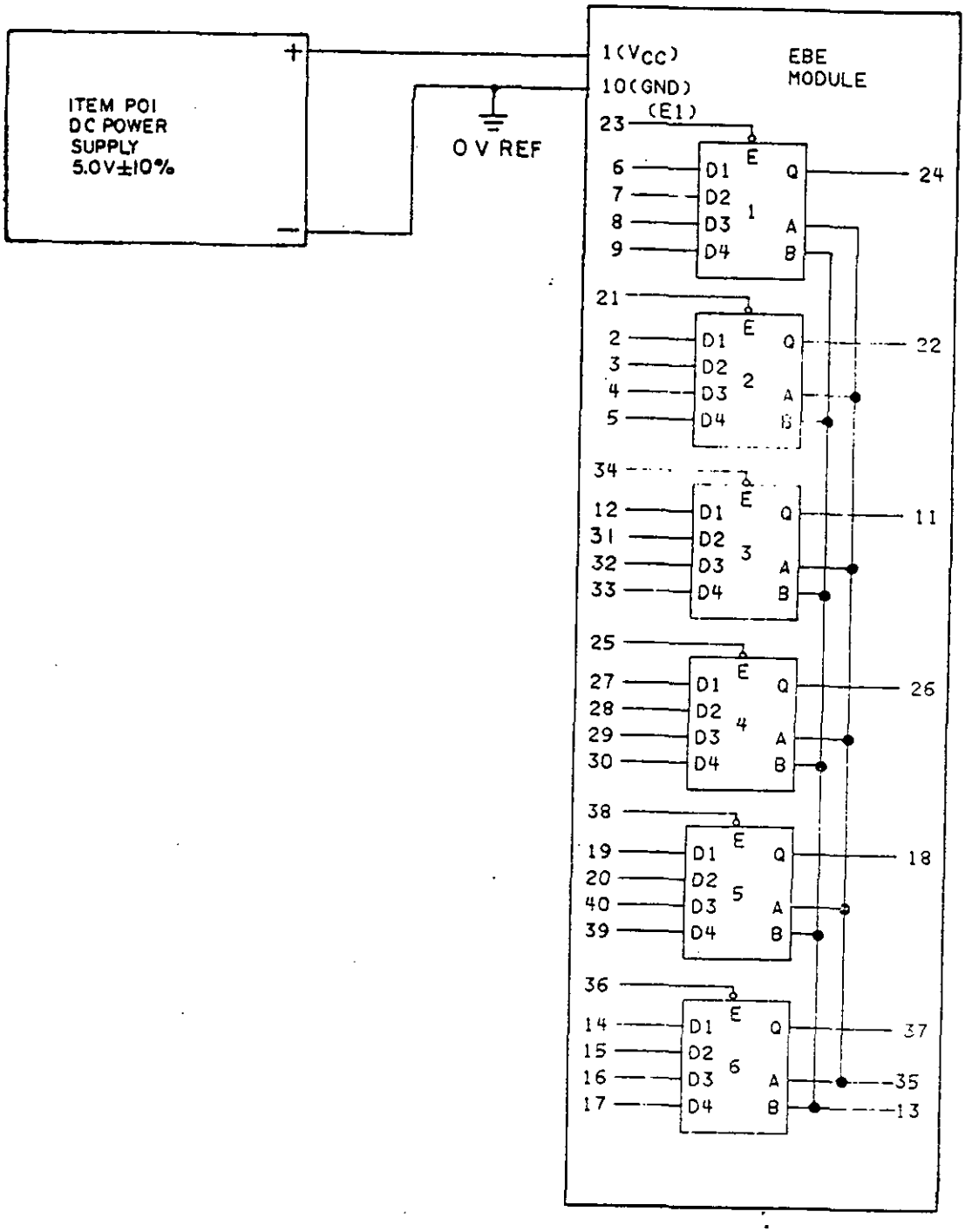


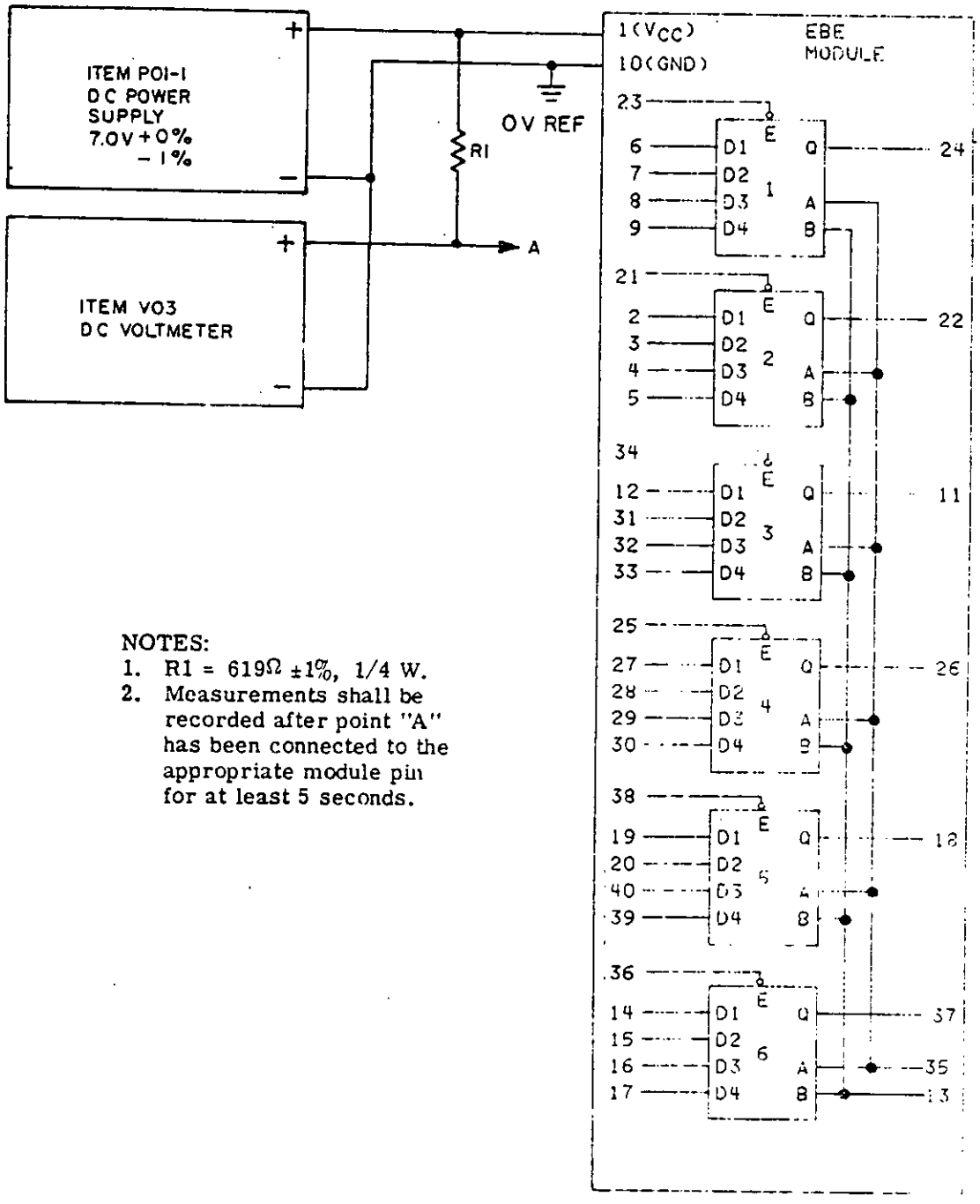
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1Q V_{OUT} (low-loaded)	24	0.01	0.80	V	5
A2	2Q	22	↓	↓	↓	↓
A3	3Q	11				
A4	4Q	26				
A5	5Q	18				
A6	6Q	37				

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NOTES:
1. R1 = 619Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	B Input breakdown current, select	13	Open	Open	0	200	µA	6
B2	A Input breakdown current, select	35	Open	Open		200	µA	
B3	1E Input breakdown current, enable	23	13,35	21,34,25,38,36		1	mA	
B4	2E	21		23,34,25,38,36				
B5	3E	34		23,21,25,38,36				
B6	4E	25		23,21,34,38,36				
B7	5E	38		23,21,34,25,36				
B8	6E	36		23,21,34,25,38				
B9	1D1 Input breakdown current, data	6		Open				
B10	2D1	2						
B11	3D1	12						
B12	4D1	27						
B13	5D1	19						
B14	6D1	14						
B15	1D2	7	35	13				
B16	2D2	3						
B17	3D2	31						
B18	4D2	28						
B19	5D2	20						
B20	6D2	15						
B21	1D3	8	13	35				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B22	2D3 Input breakdown current, data	4	13	35	0	1	mA	6
B23	3D3	32						
B24	4D3	29						
B25	5D3	40						
B26	6D3	16						
B27	1D4	9	Open	13,35				
B28	2D4	5						
B29	3D4	33						
B30	4D4	30						
B31	5D4	39						
B32	6D4	17						

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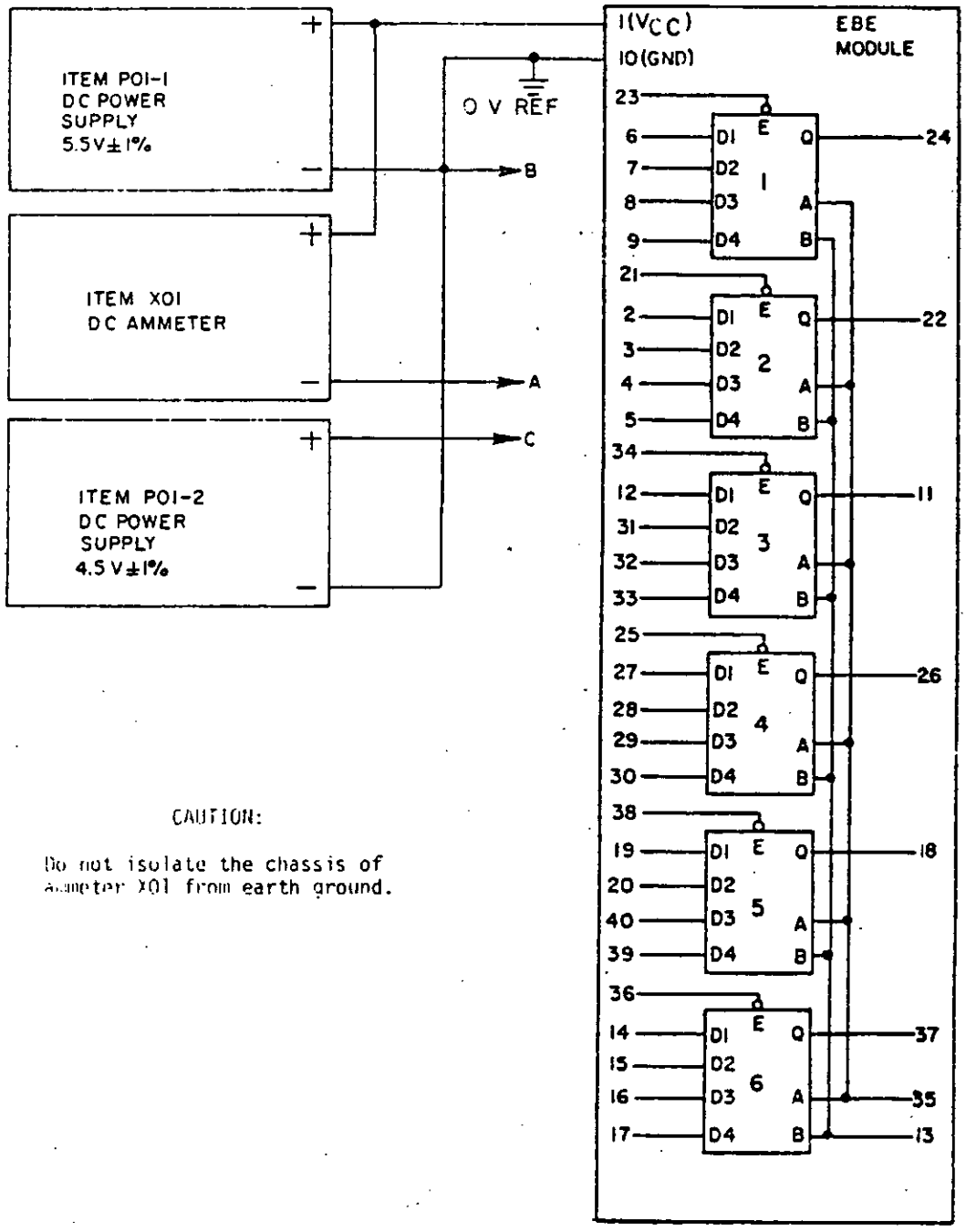


FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

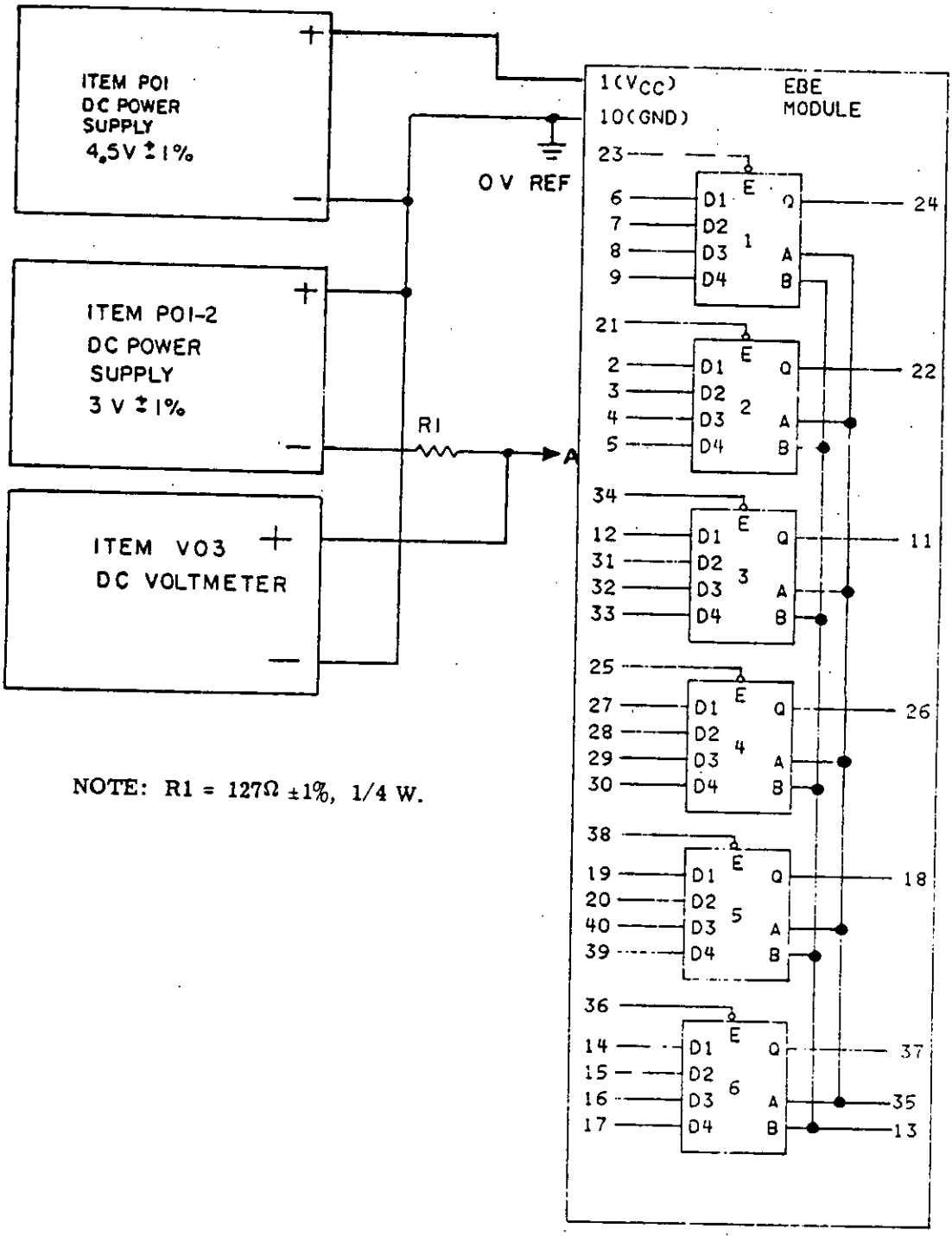
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	1E Input clamp diode voltage	23	-0.4	-1.5	V	7
C2	2E	21				
C3	3E	34				
C4	4E	25				
C5	5E	38				
C6	6E	36				
C7	1D1	6				
C8	1D2	7				
C9	1D3	8				
C10	1D4	9				
C11	2D1	2				
C12	2D2	3				
C13	2D3	4				
C14	2D4	5				
C15	3D1	12				
C16	3D2	31				
C17	3D3	32				
C18	3D4	33				
C19	4D1	27				
C20	4D2	28				
C21	4D3	29				
C22	4D4	30				
C23	5D1	19				
C24	5D2	20				

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TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C25	5D3 Input clamp diode voltage	40	-0.4	-1.5	V	7
C26	5D4	39	↓	↓	↓	↓
C27	6D1	14				
C28	6D2	15				
C29	6D3	16				
C30	6D4	17				

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NOTE: R1 = 127Ω ±1%, 1/4 W.

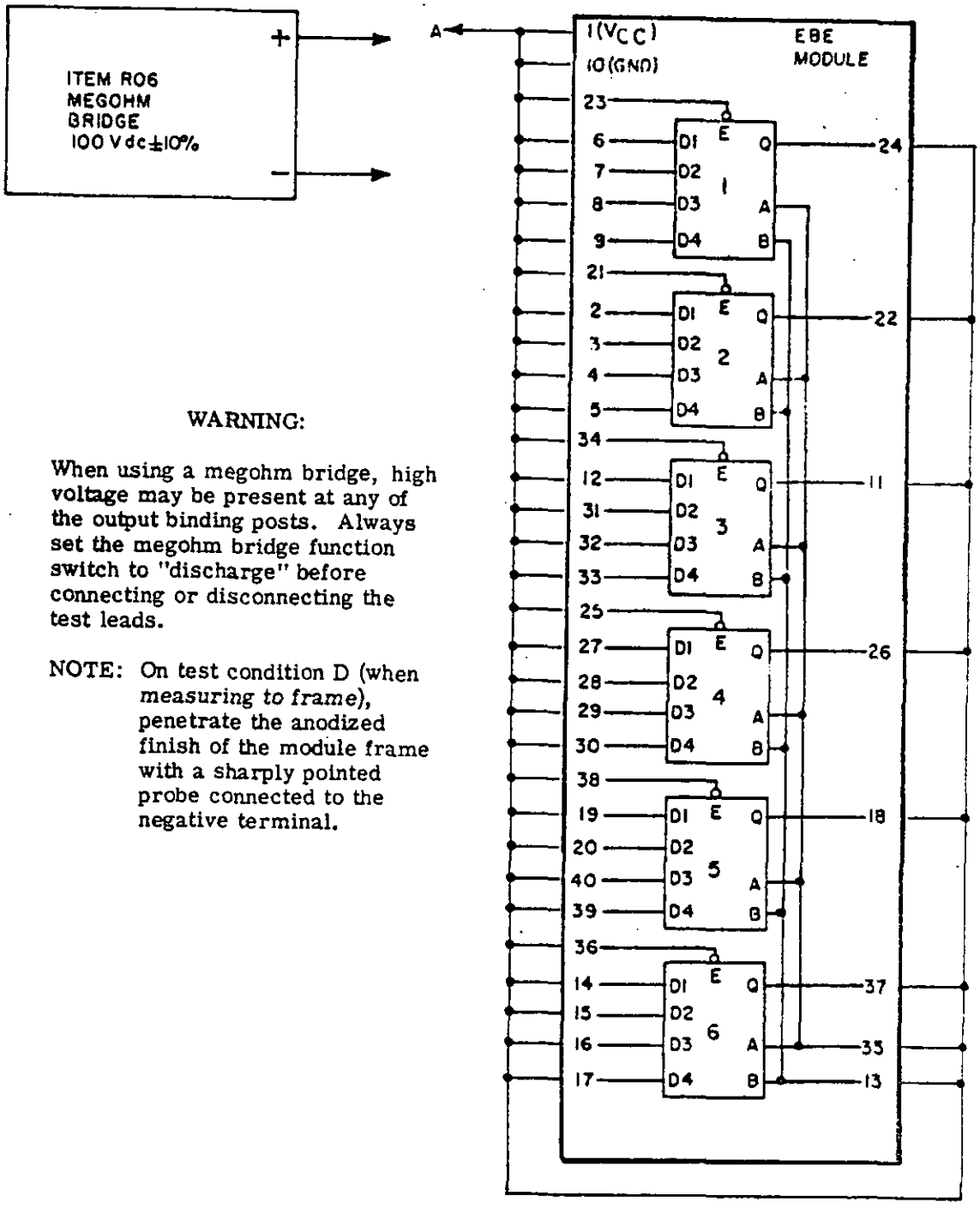
FIGURE 7. Input clamp diode voltage test circuit for table V.

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TABLE VI. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		Positive to pin	Negative to pin	Min	Max		
D	Isolation, frame	A	Frame	10	∞	MΩ	8

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition D (when *measuring to frame*), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 8. Isolation test circuit for table VI.

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TABLE VII. Static output voltage (low-level and high-level) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	1Q Static output voltage, low level	6,7,8	Open	Open	23,13, 35,9	24	0.0	0.4	V	9
E2	1Q	6,7,9		13	23,35, 8					
E3	1Q	6,8,9		35	23,13, 7					
E4	1Q	7,8,9		13,35	23,6					
E5	2Q	2,3,4		Open	21,5, 13,35	22				
E6	2Q	2,3,5		13	21,4, 35					
E7	2Q	2,4,5		35	21,3, 13					
E8	2Q	3,4,5		13,35	21,2					
E9	3Q	12,31, 32		Open	34,33 13,35	11				
E10	3Q	31,33, 12		13	34,32 35					
E11	3Q	32,33, 12		35	34,31 13					
E12	3Q	32,33, 31		13,35	34,12					
E13	4Q	28,29, 27		Open	25,30 13,35	26				
E14	4Q	28,30, 27		13	25,29 35					
E15	4Q	29,30, 27		35	25,28 13					

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TABLE VII. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E16	4Q Static output voltage, low level	29,30,28	Open	13,35	25,27	26	0.0	0.4	V	9
E17	5Q	20,40,19		Open	38,39,13,35	18				
E18	5Q	20,39,19		13	38,40,35					
E19	5Q	39,40,19		35	38,20,13					
E20	5Q	39,40,20		13,35	38,19					
E21	6Q	15,16,14		Open	36,17,13,35	37				
E22	6Q	15,17,14		13	36,16,35					
E23	6Q	16,17,14		35	36,15,13					
E24	6Q	16,17,15		13,35	36,14					
E25	1Q	35,6,13		23	Open	24				
E26	2Q	13,35,2		21		22				
E27	3Q	13,35,12		34		11				
E28	4Q	27,35,13		25		26				
E29	5Q	19,35,13		38		18				
E30	6Q	14,35,13		36		37				

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TABLE VII. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	1Q Static output voltage, high level	Open	7,8, 23,6	9	13,35	24	2.4	4.5	V	9
F2	1Q		6,9, 23,7	8,13	35					
F3	1Q		6,9, 23,8	7,35	13					
F4	1Q		7,8, 23,9	6,13, 35	Open					
F5	2Q		3,4, 21,2	5	13,35	22				
F6	2Q		2,5, 21,3	4,13	35					
F7	2Q		2,5, 21,4	3,35	13					
F8	2Q		3,4, 21,5	2,13, 35	Open					
F9	3Q		31,32, 34,12	33	13,35	11				
F10	3Q		12,33, 34,31	32,13	35					
F11	3Q		12,33, 34,32	31,35	13					
F12	3Q		31,32, 34,33	12,13, 35	Open					
F13	4Q		28,29, 25,27	30	13,35	26				
F14	4Q		27,30, 25,28	29,13	35	26				
F15	4Q		27,30, 25,29	28,29 35	13	26				

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TABLE VII. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F16	4Q Static output voltage, high level	Open	28,29, 25,30	27,13, 35	Open	26	2.4	4.5	V	9
F17	5Q		20,40, 38,19	39	13,35	18				
F18	5Q		19,39, 38,20	40,13	35					
F19	5Q		19,39, 38,40	20,35	13					
F20	5Q		20,40, 38,39	19,13, 35	Open					
F21	6Q		15,16, 36,14	17	13,35	37				
F22	6Q		14,17, 36,15	16,13	35					
F23	6Q		14,17, 36,16	15,35	13					
F24	6Q		15,16, 36,17	14,13, 35	Open					

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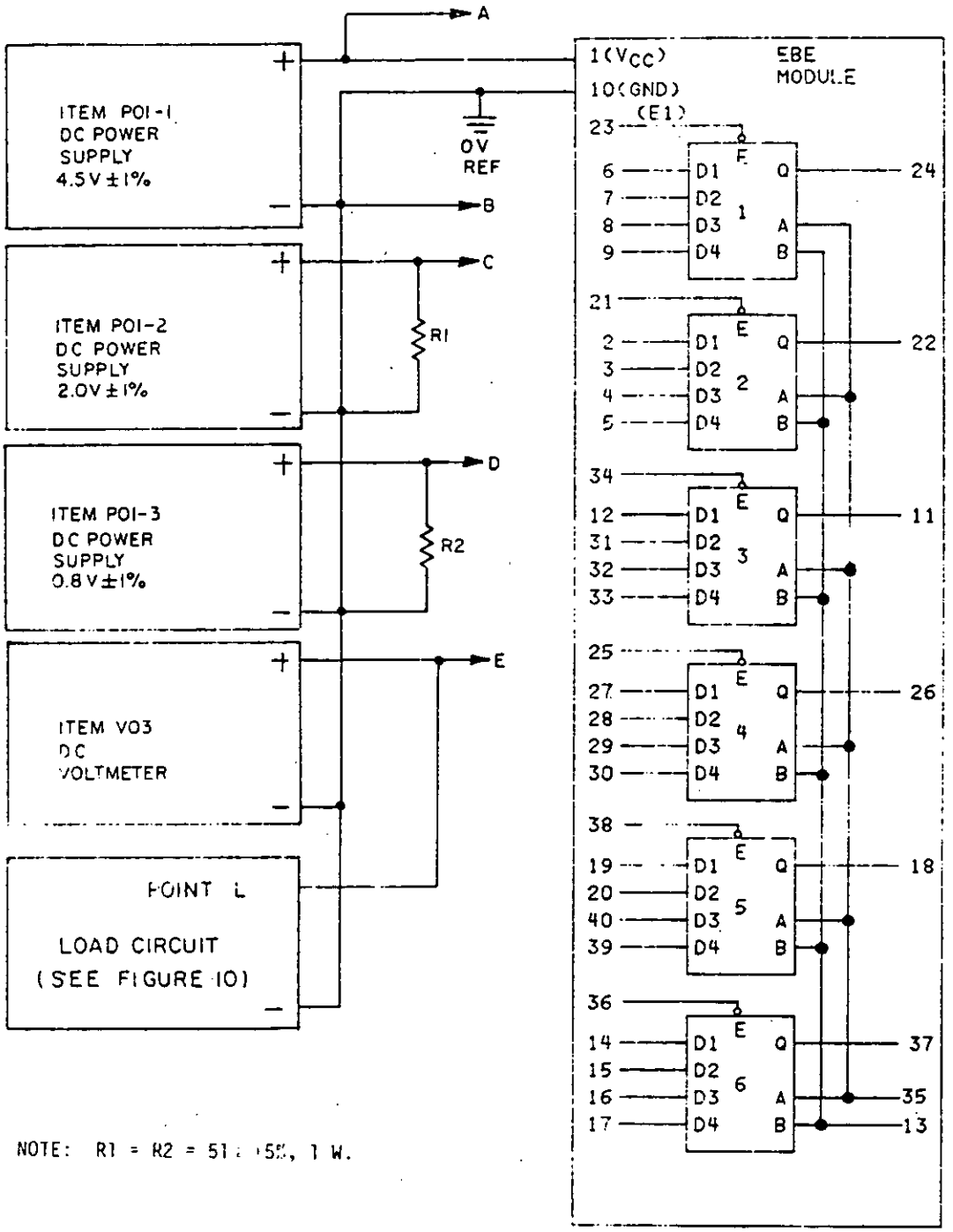
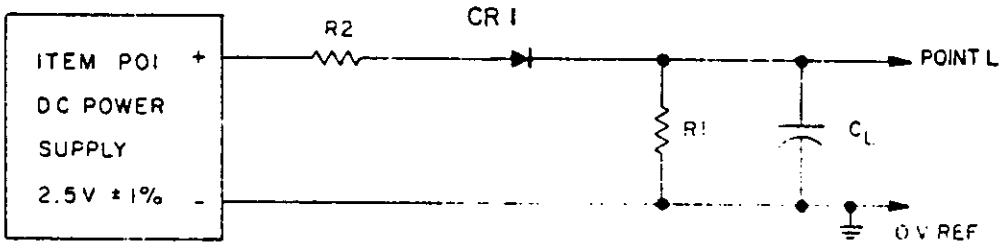


FIGURE 9. Static output voltage (low-level and high-level) test circuit for table VII.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 0.04 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 8.0 mA ±5% out of point L with point L held at 0.4 V.
4. For test conditions N1-N42 and P1-P42:
   C_L = 40 ±5 pF (including probe and parasitic capacitance).
5. For test conditions N43-N66 and P43-P66:
   C_L = 140 ±5 pF (including probe and parasitic capacitance).
6. For test conditions R1-R12:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
7. Capacitor C_L is not required for dc electrical test conditions E and F.
8. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
9. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 10. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	B Input current, select, low level	35,23, 21,34, 25,38,36	Open	13	Open	0.05	0.40	mA	11
G2	A Input current, select, low level	13,23, 21,34, 25,38,36		35	Open	0.05	0.40		
G3	1D1 Input current, data,low level	23		6	7,8,9, 13,35	0.1	0.9		
G4	1D2	23,13		7	6,8, 9,35				
G5	1D3	23,35		8	6,7, 9,13				
G6	1D4	23,13,35		9	6,7,8				
G7	2D1	21		2	3,4,5, 13,35				
G8	2D2	21,13		3	2,4, 5,35				
G9	2D3	21,35		4	2,3, 5,13				
G10	2D4	21,13,35		5	2,3,4				
G11	3D1	34		12	13,31, 32,33, 35				
G12	3D2	34,13		31	12,32, 33,35				
G13	3D3	34,35		32	12,31, 33,13				
G14	3D4	34,13,35		33	12,31, 32				

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G15	4D1 Input current, data, low level	25	Open	27	13,28, 29,30, 35	0.1	0.9	mA	11
G16	4D2	25,13		28	27,29, 30,35				
G17	4D3	25,35		29	13,27, 28,30				
G18	4D4	25,13, 35		30	27,28, 29				
G19	5D1	38		19	13,20, 40,39, 35				
G20	5D2	38,13		20	19,40, 39,35				
G21	5D3	38,35		40	13,19, 20,39				
G22	5D4	38,13, 35		39	19,20, 40				
G23	6D1	36		14	13,15, 16,17, 35				
G24	6D2	36,13		15	14,16, 17,35				
G25	6D3	36,35		16	13,14, 15,17				
G26	6D4	36,13, 35		17	14,15, 16				
G27	1E Input current, enable, low level	6,7,8,9		23	13,35				

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parame:	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G28	2E Input current, enable, low level	2,3,4,5	Open	21	13,35	0.1	0.9	mA	11
G29	3E	12,31, 32,33		34					
G30	4E	27,28, 29,30		25					
G31	5E	19,20, 40,39		38					
G32	6E	14,15, 16,17		36					

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	A Input current, select, high level	Open	35	Open	23,21, 34,13, 25,38, 36	0	20	µA	11
H2	B Input current, select, high level	Open	13		23,21, 34,35, 25,38, 36				
H3	1D1 Input current, data, high level	7,8,9, 13,35	6		23				
H4	1D2	6,8,9, 35	7		13,23				
H5	1D3	6,7,9, 13	8		35,23				
H6	1D4	6,7,8	9		13,35, 23				
H7	2D1	3,4,5, 13,35	2		21				
H8	2D2	2,4,5, 35	3		13,21				
H9	2D3	2,3,5, 13	4		35,21				
H10	2D4	2,3,4	5		13,35, 21				
H11	3D1	31,32, 33,13, 35	12		34				
H12	3D2	12,32, 33,35	31		13,34				
H13	3D3	12,31, 33,13	32		35,34				

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TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H14	3D4 Input current, data, high level	12,31, 32	33	Open	13,35, 34	0	20	µA	11
H15	4D1	28,29, 30,13, 35	27		25				
H16	4D2	27,29, 30,35	28		13,25				
H17	4D3	27,28, 30,13	29		35,25				
H18	4D4	27,28, 29	30		13,35, 25				
H19	5D1	20,40, 39,13, 35	19		38				
H20	5D2	19,40, 39,35	20		13,38				
H21	5D3	19,20, 39,13	40		35,38				
H22	5D4	19,20, 40	39		13,35, 38				
H23	6D1	15,16, 17,13, 35	14		36				
H24	6D2	14,16, 17,35	15		13,36				
H25	6D3	14,15, 17,13	16		35,36				
H26	6D4	14,15, 16	17		13,35, 36				

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TABLE IX. Input current (high-level) test - Continued.

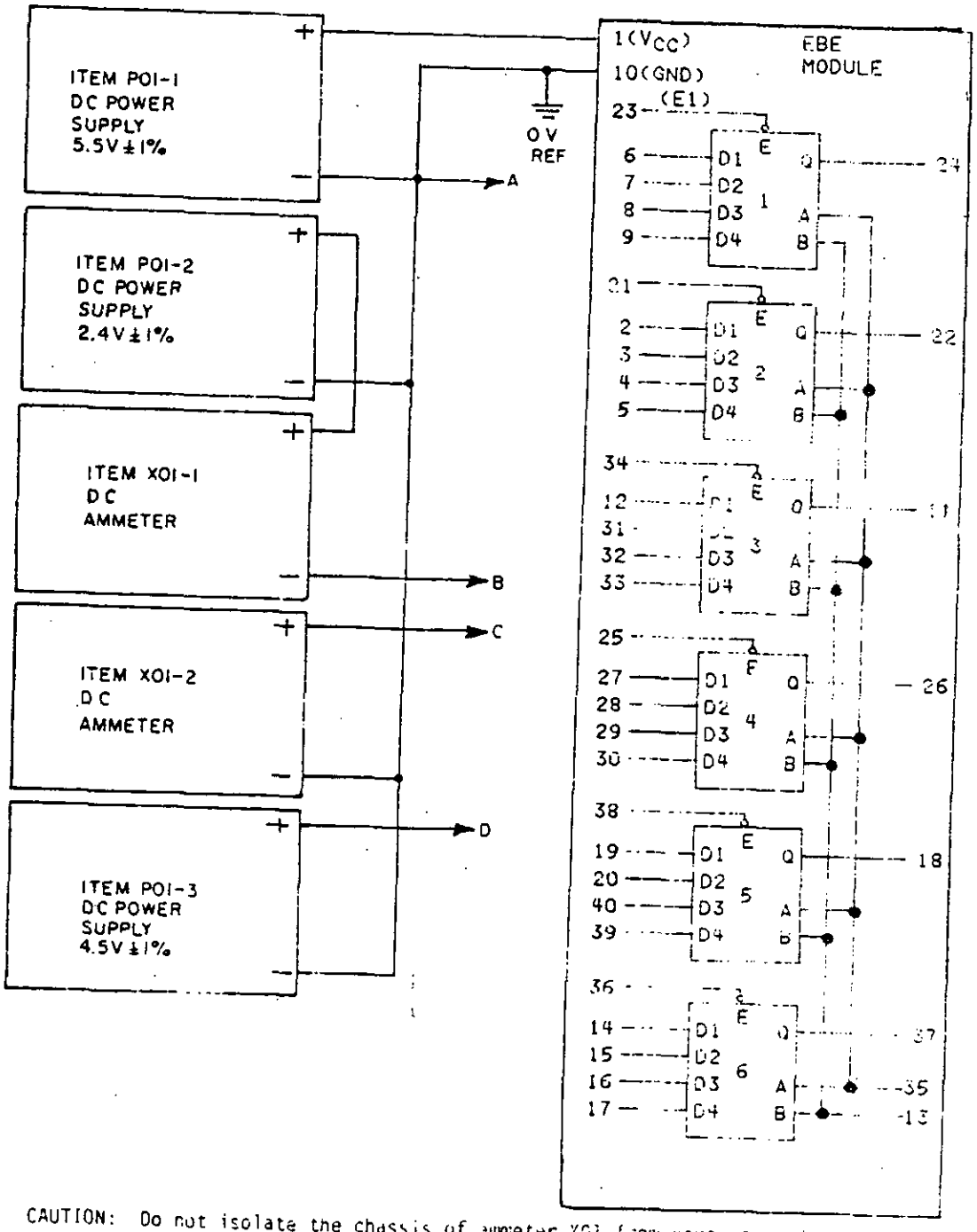
T. · con· tion	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H27	1E Input current, enable, high level	6,7,8,9, 13,35	23	Open	Open	0	20	µA	11
H2·	2E	2,3,4,5, 13,35	21						
H3	3E	12,31, 32,33, 13,35	34						
H3	4E	27,28, 29,30, 13,35	25						
E31	5E	19,20, 40,39, 13,35	38						
H32	6E	14,15, 16,17, 13,35	36						

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1Q Output short circuit current	23,13, 35	Open	24	9	10	28	mA	11
J2	2Q	21,13, 35		22	5				
J3	3Q	34,13, 35		11	33				
J4	4Q	25,13, 35		26	30				
J5	5Q	38,13, 35		18	39				
J6	6Q	36,13, 35		37	17				

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CAUTION: Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:

1. Since the ammeters are not used simultaneously, only one de[illegible] required. Use ammeter XOI-2 for tables VIII and X. Use ammeter XOI-1 [illegible] table IX.
2. With point "B" disconnected, ammeter XOI-1 should indicate [illegible] the 1 μA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX and X.

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TABLE XI. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current	5	81	mA	12

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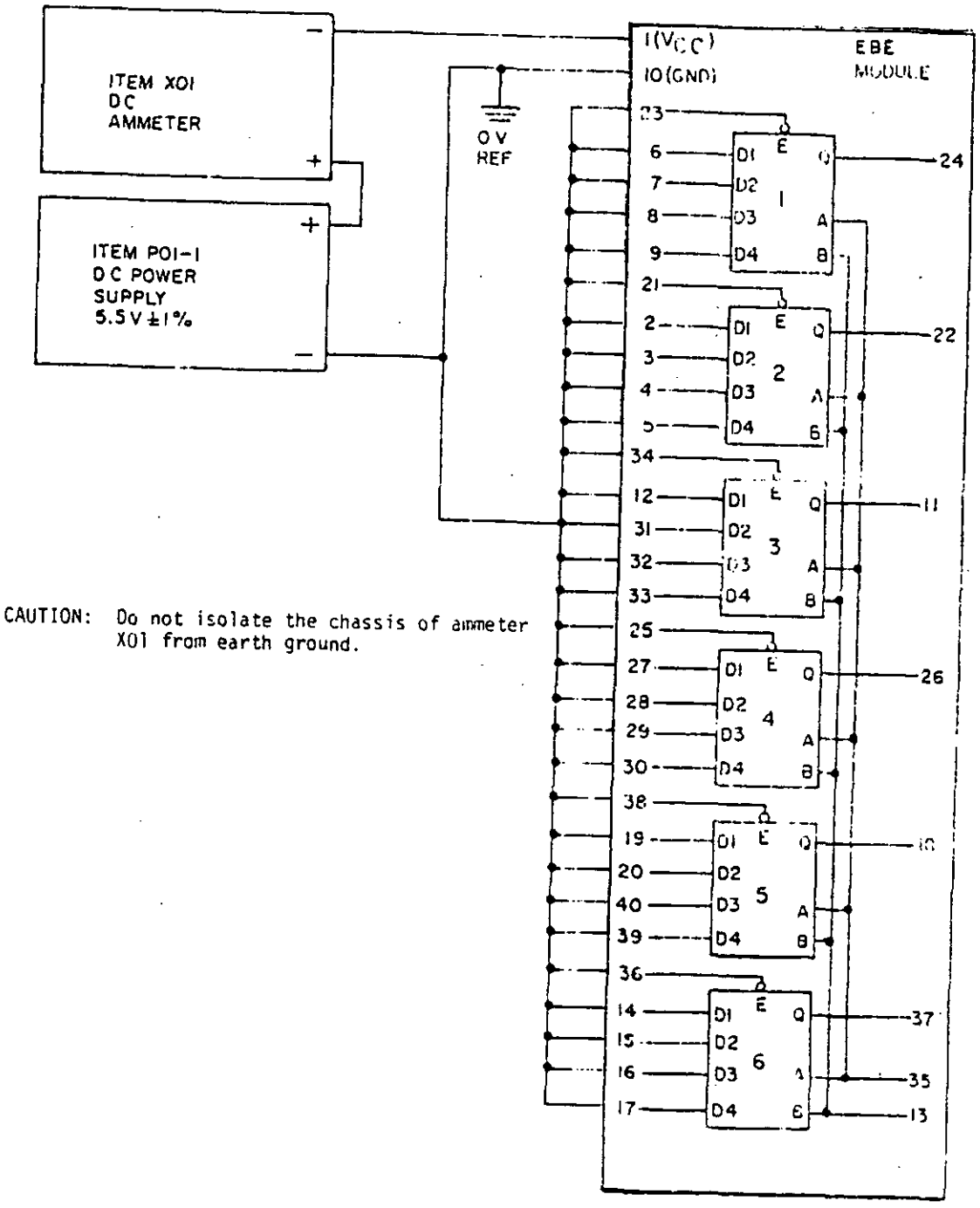


FIGURE 12. Power supply current test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1Q Transition time (t_{TLH}), C_L = 40 pF	35,13	9	23	24	2	25	ns	13,14
L2	2Q		5	21	22				
L3	3Q		33	34	11				
L4	4Q		30	25	26				
L5	5Q		39	38	18				
L6	6Q		17	36	37				
L7	1Q Transition time (t_{TLH}), C_L = 140 pF		9	23	24	5	45		
L8	2Q		5	21	22				
L9	3Q		33	34	11				
L10	4Q		30	25	26				
L11	5Q		39	38	18				
L12	6Q		17	36	37				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Q Transition time (t_{THL}), C_L = 40 pF	35,13	9	23	24	1	20	ns	13,14
M2	2Q		5	21	22				
M3	3Q		33	34	11				
M4	4Q		30	25	26				
M5	5Q		39	38	18				
M6	6Q		17	36	37				
M7	1Q Transition time (t_{THL}), C_L = 140 pF		9	23	24	2	30		
M8	2Q		5	21	22				
M9	3Q		33	34	11				
M10	4Q		30	25	26				
M11	5Q		39	38	18				
M12	6Q		17	36	37				

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TABLE IV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1D4 to 1Q Propagation delay time (t_{PLH}), C_L = 40 pF	35,13, 23	Open	9	24	5	40	ns	13,14
N2	2D4 to 2Q	35,13, 21		5	22				
N3	3D4 to 3Q	35,13, 34		33	11				
N4	4D4 to 4Q	35,13, 25		30	26				
N5	5D4 to 5Q	35,13, 38		39	18				
N6	6D4 to 6Q	35,13, 36		17	37				
N7	1D3 to 1Q	35,23	13	8	24				
N8	1D2 to 1Q	13,23	35	7	24				
N9	1D1 to 1Q	23	13,35	6	24				
N10	2D3 to 2Q	35,21	13	4	22				
N11	2D2 to 2Q	13,21	35	3	22				
N12	2D1 to 2Q	21	13,35	2	22				
N13	3D3 to 3Q	35,34	13	32	11				
N14	3D2 to 3Q	13,34	35	31	11				
N15	3D1 to 3Q	34	13,35	12	11				
N16	4D3 to 4Q	35,25	13	29	26				
N17	4D2 to 4Q	13,25	35	28	26				
N18	4D1 to 4Q	25	13,35	27	26				
N19	5D3 to 5Q	35,38	13	40	18				
N20	5D2 to 5Q	13,38	35	20	18				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N21	5D1 to 5Q Propagation delay time	38	13,35	19	18	5	40	ns	13,14
N22	6D3 to 6Q (t_{PLH}) C_L = 40 pF	35,36	13	16	37				
N23	6D2 to 6Q	13,36	35	15	37				
N24	6D1 to 6Q	36	13,35	14	37				
N25	A to 1Q	7,13,23	9	35	24		120		
N26	A to 2Q	3,13,21	5		22				
N27	A to 3Q	31,13,34	33		11				
N28	A to 4Q	28,13,25	30		26				
N29	A to 5Q	20,13,38	39		18				
N30	A to 6Q	15,13,36	17		37				
N31	B to 1Q	8,23,35	9	13	24				
N32	B to 2Q	4,21,35	5		22				
N33	B to 3Q	32,34,35	33		11				
N34	B to 4Q	25,29,35	30		26				
N35	B to 5Q	35,38,40	39		18				
N36	B to 6Q	16,35,36	17		37				
N37	B to 1Q	35,13	9	23	24		66		

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N38	2E to 2Q Propagation delay time	35,13	5	21	22	5	66	ns	13,14
N39	3E to 3Q (t_{PLH}), C_L = 40 pF		33	34	11				
N40	4E to 4Q		30	25	26				
N41	5E to 5Q		39	38	18				
N42	6E to 6Q		17	36	37				
N43	1D4 to 1Q Propagation delay time (t_{PLH}), C_L = 140 pF	35,13, 23	Open	9	24		60		
N44	2D4 to 2Q	35,13, 21		5	22				
N45	3D4 to 3Q	35,13, 34		33	11				
N46	4D4 to 4Q	35,13, 25		30	26				
N47	5D4 to 5Q	35,13, 38		39	18				
N48	6D4 to 6Q	35,13, 36		17	37				
N49	A to 1Q	7,13,23	9	35	24		140		
N50	A to 2Q	3,13,21	5		22				
N51	A to 3Q	31,13, 34	33		11				
N52	A to 4Q	28,13, 25	30		26				
N53	A to 5Q	20,13, 38	39		18				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N54	A to 6Q Propagation delay time (t_{PLH}), C_L = 140 pF	15,13, 36	17	35	37	5	140	ns	13,14
N55	B to 1Q	8,23,35	9	13	24				
N56	B to 2Q	4,21,35	5		22				
N57	B to 3Q	32,34, 35	33		11				
N58	B to 4Q	29,25, 35	30		26				
N59	B to 5Q	35,38, 40	39		18				
N60	B to 6Q	16,35, 36	17		37				
N61	1E to 1Q	35,13	9	23	24		86		
N62	2E to 2Q		5	21	22				
N63	3E to 3Q		33	34	11				
N64	4E to 4Q		30	25	26				
N65	5E to 5Q		39	38	18				
N66	6E to 6Q		17	36	37				

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TABLE IV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Te fi
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1D4 to 1Q Propagation delay time (t_{PHL}), C_L = 40 pF	35,13, 23	Open	9	24	5	50	ns	1
P2	2D4 to 2Q	35,13, 21		5	22				
P3	3D4 to 3Q	35,13, 34		33	11				
P4	4D4 to 4Q	35,13, 25		30	26				
P5	5D4 to 5Q	35,13, 38		39	18				
P6	6D4 to 6Q	35,13, 36		17	37				
P7	1D3 to 1Q	35,23	13	8	24				
P8	1D2 to 1Q	13,23	35	7	24				
P9	1D1 to 1Q	23	13,35	6	24				
P10	2D3 to 2Q	35,21	13	4	22				
P11	2D2 to 2Q	13,21	35	3	22				
P12	2D1 to 2Q	21	13,35	2	22				
P13	3D3 to 3Q	35,34	13	32	11				
P14	3D2 to 3Q	13,34	35	31	11				
P15	3D1 to 3Q	34	13,35	12	11				
P16	4D3 to 4Q	35,25	13	29	26				
P17	4D2 to 4Q	13,25	35	28	26				
P18	4D1 to 4Q	25	13,35	27	26				
P19	5D3 to 5Q	35,38	13	40	18				
P20	5D2 to 5Q	13,38	35	20	18				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P21	5D1 to 5Q Propagation delay time	38	13,35	19	18	5	50	ns	13,14
P22	6D3 to 6Q (t_{PHL}), C_L = 40 pF	35,36	13	16	37				
P23	6D2 to 6Q	13,36	35	15	37				
P24	6D1 to 6Q	36	13,35	14	37				
P25	A to 2Q	7,13, 23	9	35	24		120		
P26	A to 4Q	3,13,21	5		22				
P27	A to 3Q	31,13, 34	33		11				
P28	A to 4Q	28,13, 25	30		26				
P29	A to 5Q	20,13, 38	39		18				
P30	A to 7Q	15,13, 36	17		37				
P31	B to 1Q	8,23,35	9	13	24				
P32	B to 2Q	4,21,35	5		22				
P33	B to 3Q	32,34, 35	33		11				
P34	B to 4Q	25,29, 35	30		26				
P35	B to 5Q	35,38, 40	39		18				
P36	B to 6Q	16,35, 36	17		37				
P37	B to 7Q	35,13	9	23	24		50		

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° +5°C ambient Min	Initial limits 25° +5°C ambient Max	Unit	Test figure
P38	2E to 2Q Propagation delay time (t_{PHL}), C_L = 40 pF	35,13	5	21	22	3	50	ns	13,14
P39	3E to 3Q		33	34	11				
P40	4E to 4Q		30	25	26				
P41	5E to 5Q		39	38	18				
P42	6E to 6Q		17	36	37				
P43	1D4 to 1Q Propagation delay time (t_{PHL}), C_L = 140 pF	35,13, 23	Open	9	24		70		
P44	2D4 to 2Q	35,13, 21		5	22				
P45	3D4 to 3Q	35,13, 34		33	11				
P46	4D4 to 4Q	35,13, 25		30	26				
P47	5D4 to 5Q	35,13, 38		39	18				
P48	6D4 to 6Q	35,13, 36		17	37				
P49	A to 1Q	7,13,23	9	35	24		140		
P50	A to 2Q	3,13,21	5		22				
P51	A to 3Q	31,13, 34	33		11				
P52	A to 4Q	28,13, 25	30		26				
P53	A to 5Q	20,13, 38	39		18				
P54	A to 6Q	15,13, 36	17		37				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P55	B to 1Q Propagation delay time (t_{PHL}), C_L = 140 pF	8,23,35	9	13	24	5	140	ns	13,14
P56	B to 2Q	4,21,35	5		22				
P57	B to 3Q	32,34,35	33		11				
P58	B to 4Q	25,29,35	30		26				
P59	B to 5Q	35,38,40	39		18				
P60	B to 6Q	16,35,36	17		37				
P61	1E to 1Q	35,13	9	23	24		70		
P62	2E to 2Q		5	21	22				
P63	3E to 3Q		33	34	11				
P64	4E to 4Q		30	25	26				
P65	5E to 5Q		39	38	18				
P66	6E to 6Q		17	36	37				

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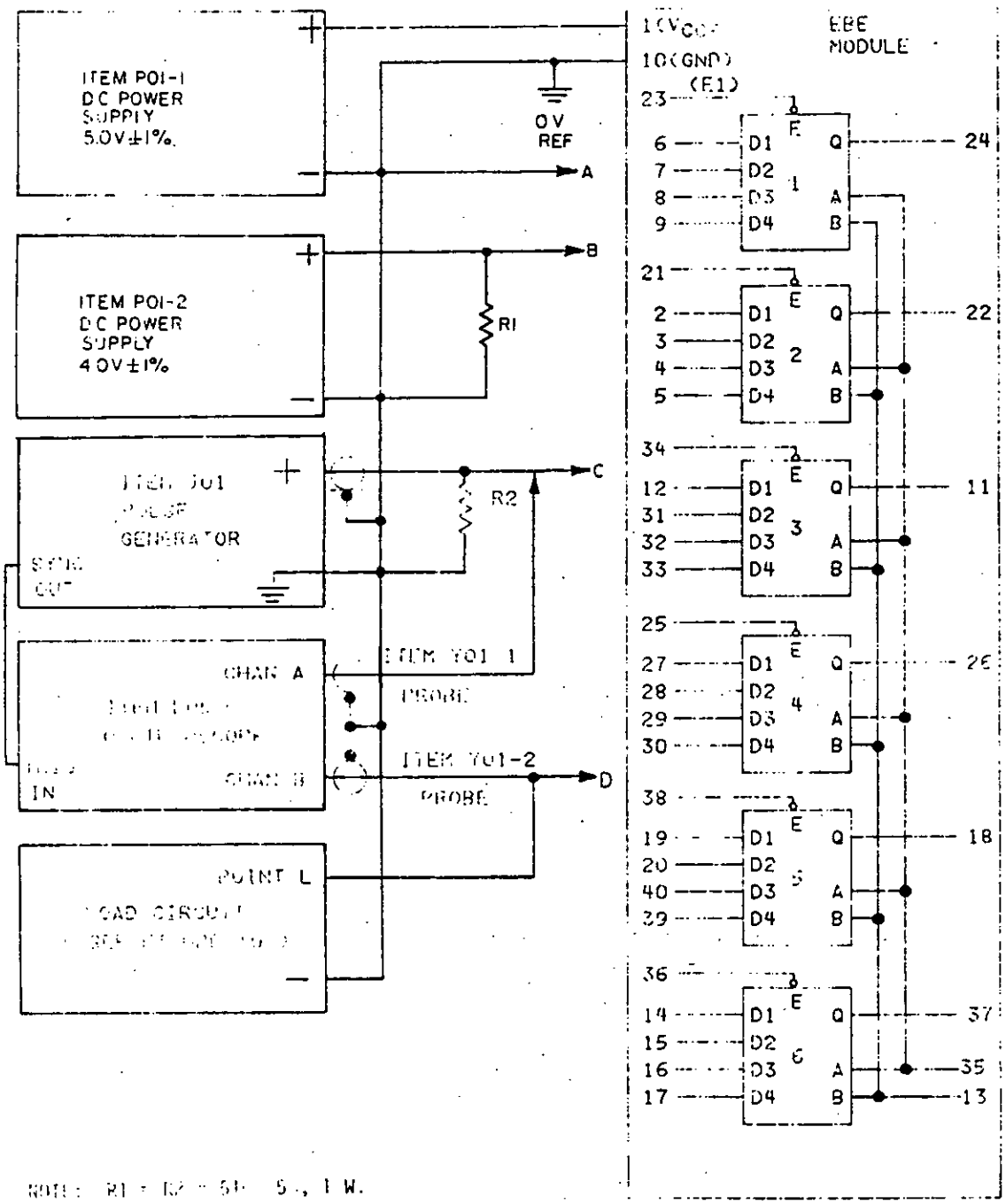
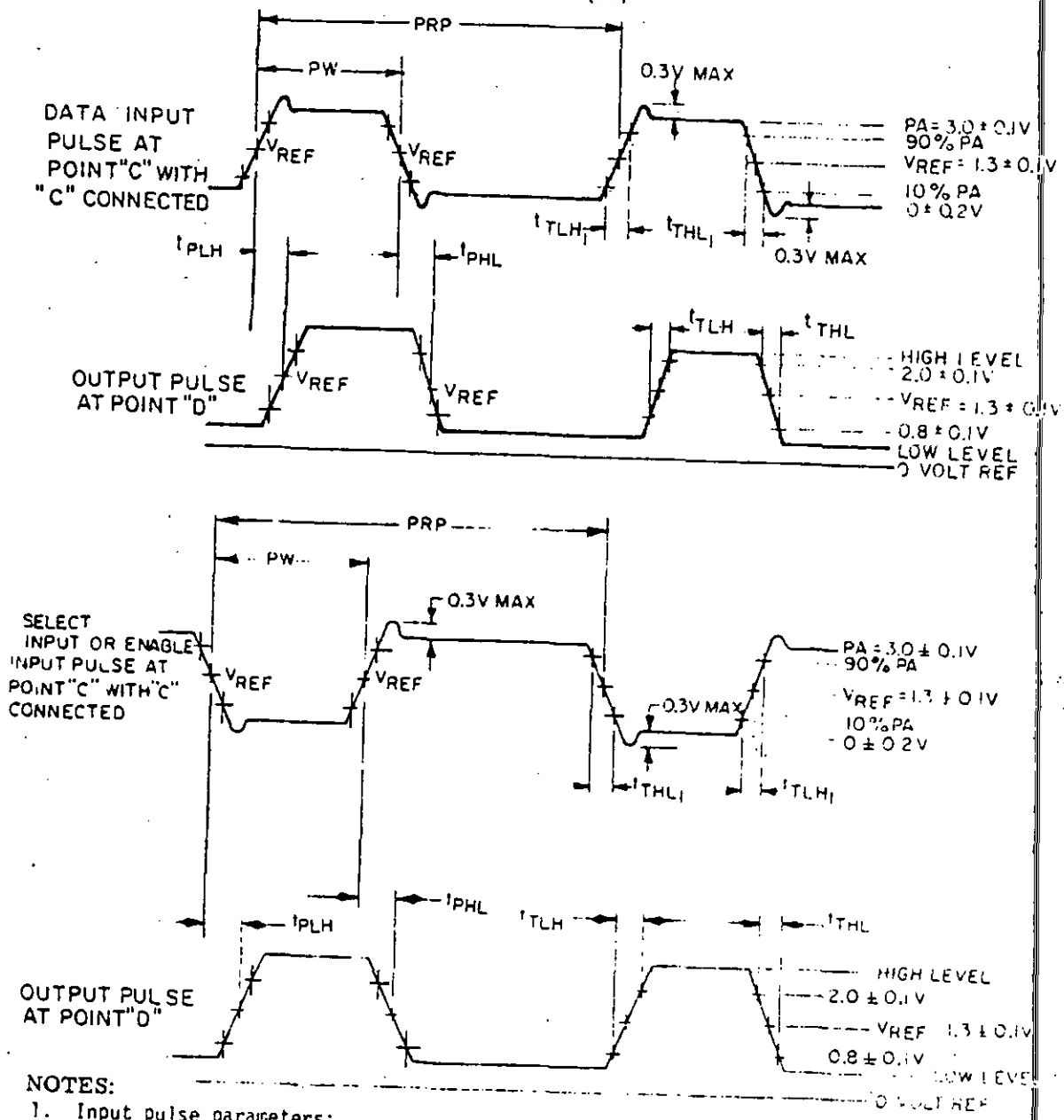


FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

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NOTES:

1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH_1} = t_{THL_1}$ = 10 ±2 ns.
2. Before making measurement, verify that the output high level is ≥ 2.1 V and the output low level is ≤ 0.4 V.
3. All voltages are referenced to ground (0 V REF).

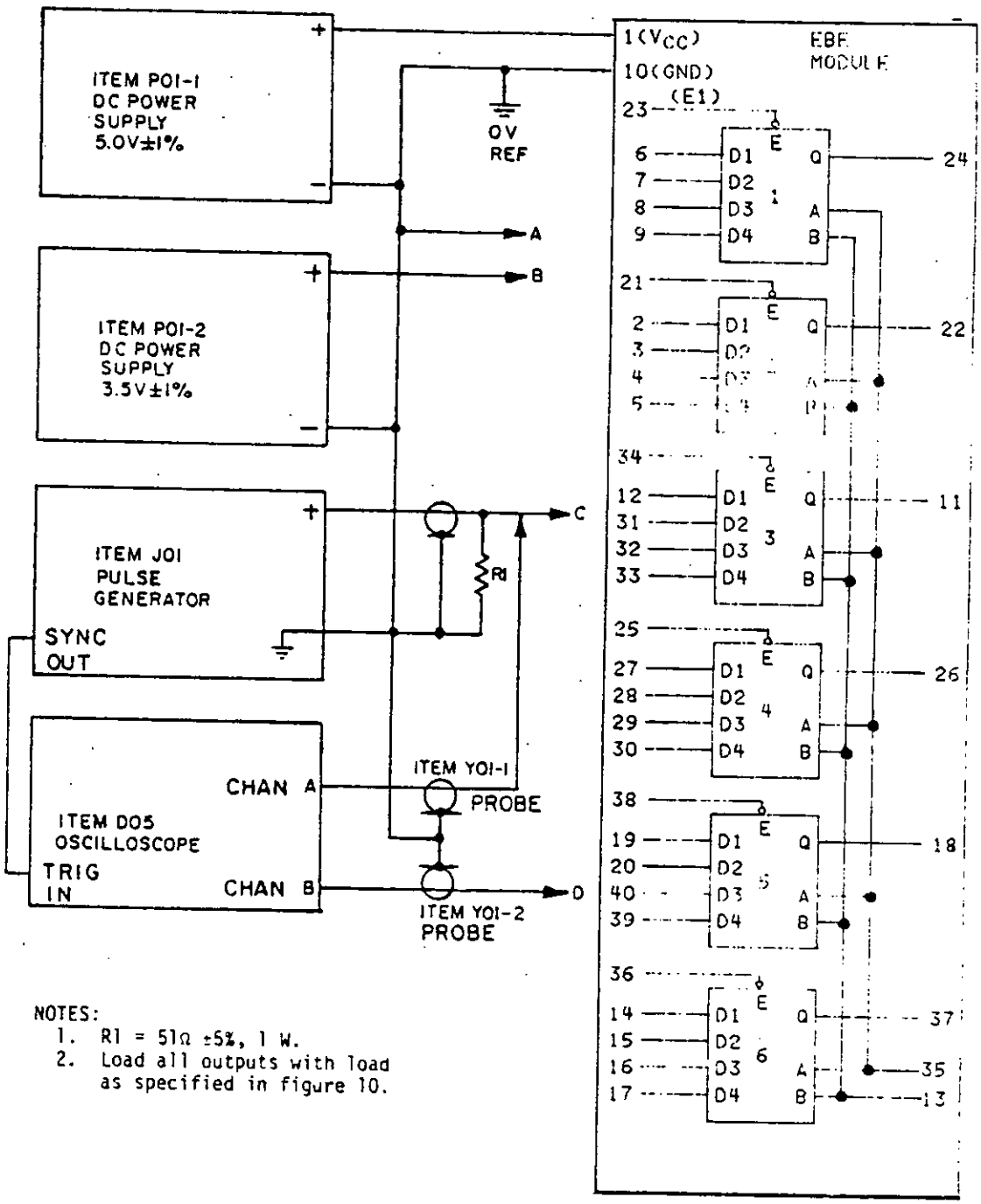
FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV, and XV.

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TABLE XVI. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Limits -55°C, +125°C, +5°C ambient Min	Limits -55°C, +125°C, +5°C ambient Max	Unit	Test figure
R1	1Q Crosstalk, low level	9,13,35,21,23,34,25,38,36	Open	5,33,30,39,17	24	0	400	mV	15,16
R2	2Q	5,13,35,21,23,34,25,38,36		9,33,30,39,17	22				
R3	3Q	33,13,35,21,23,34,25,38,36		5,9,30,39,17	11				
R4	4Q	30,13,35,21,23,34,25,38,36		5,9,33,39,17	26				
R5	5Q	39,13,35,21,23,34,25,38,36		5,9,33,30,17	18				
R6	6Q	17,13,35,21,23,34,25,38,36		5,9,33,30,39	37				
R7	1Q Crosstalk, high level	13,35,21,23,34,25,38,36	9	5,33,30,39,17	24				
R8	2Q		5	9,33,30,39,17	22				
R9	3Q		33	5,9,30,39,17	11				
R10	4Q		30	5,9,33,39,17	26				
R11	5Q		39	5,9,33,30,17	18				
R12	6Q		17	5,9,33,30,39	37				

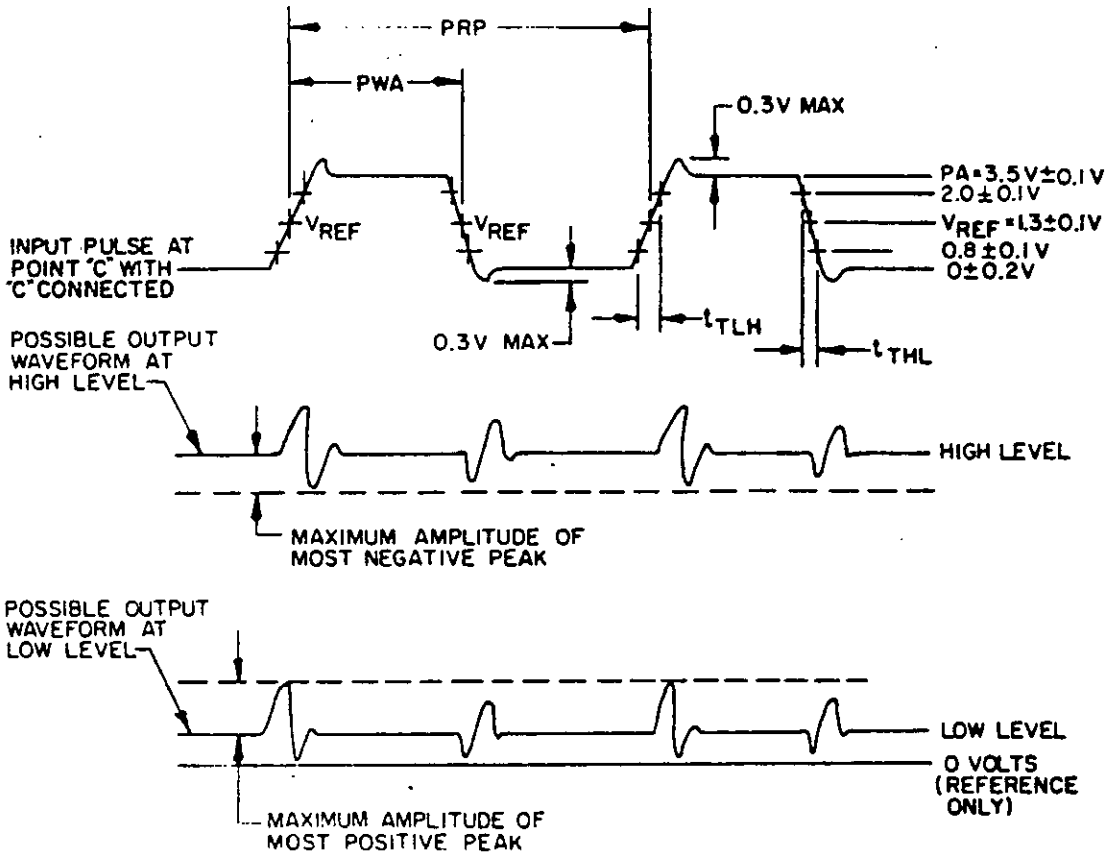
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified in figure 10.

FIGURE 15. Output crosstalk (low-level and high-level) test circuit for table XVI.

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NOTE: Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

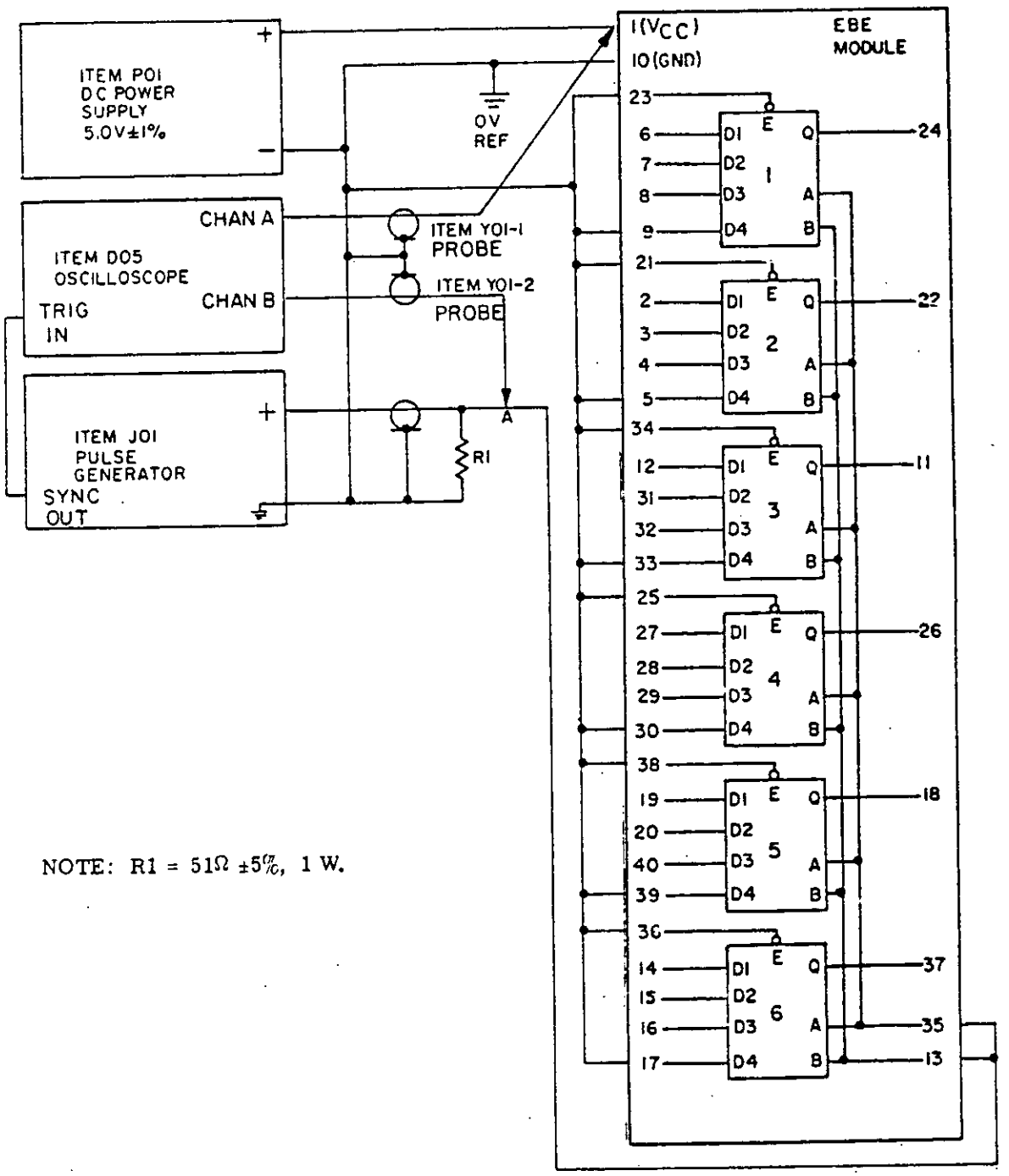
FIGURE 16. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI.

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	17,18

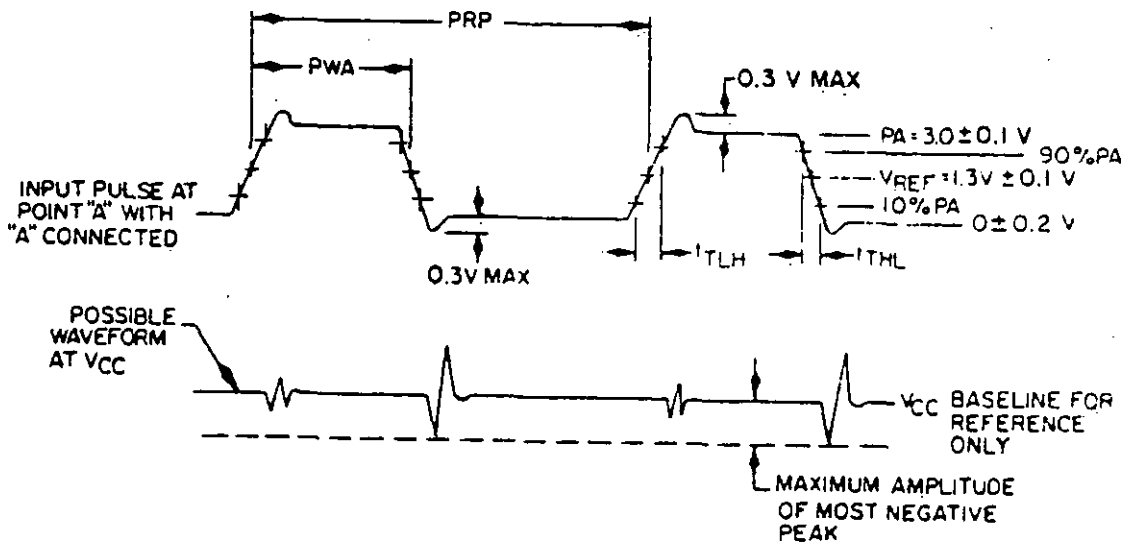
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 17. Power supply transient amplitude test circuit for table XVII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

FIGURE 18. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	19

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TABLE XIX. Input capacitance test.

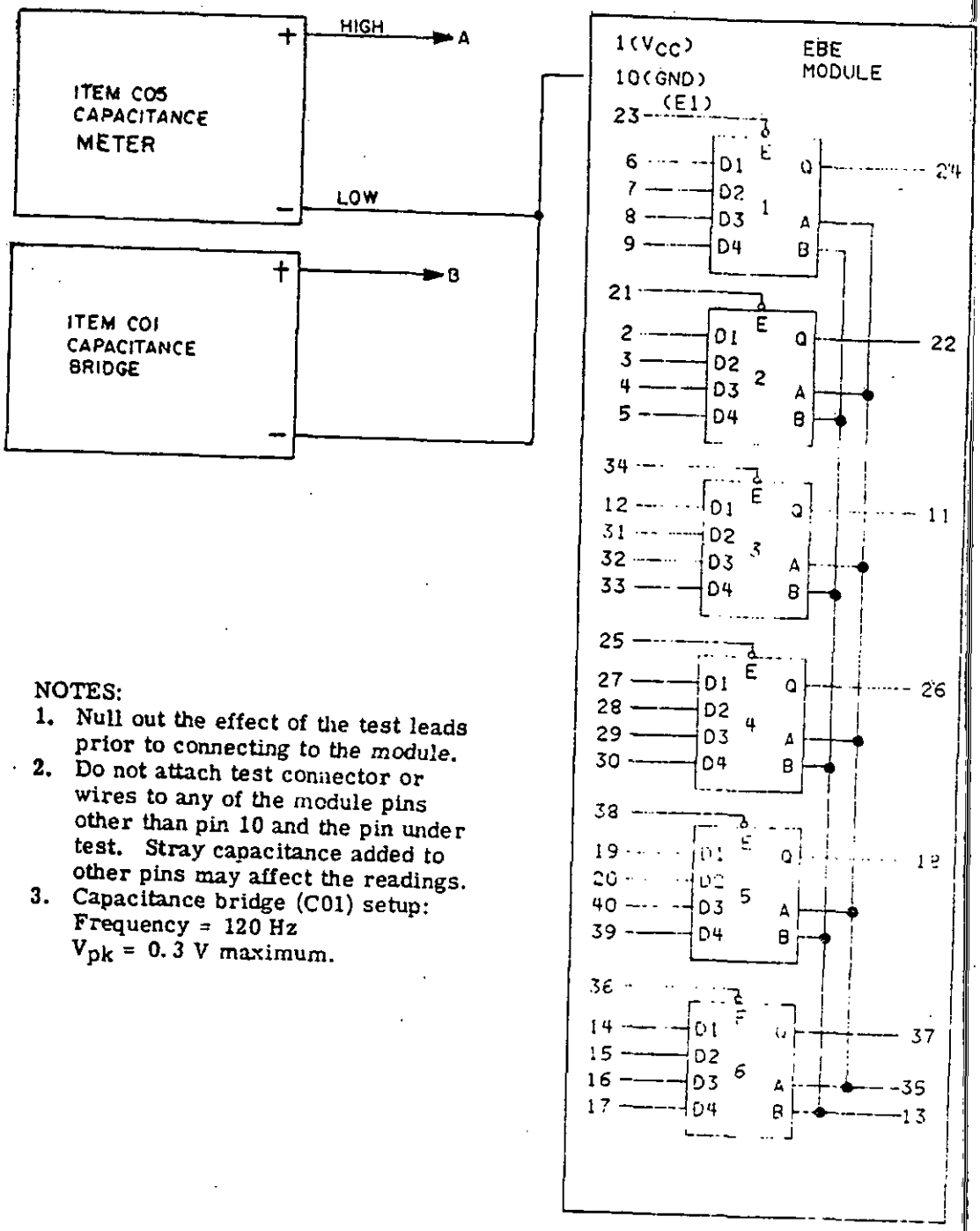
Test condition	Parameter	Connect		Initial limits 25° ±5°C - ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1D1 Input capacitance, data	6	Open	2	15	pF	19
U2	1D2	7					
U3	1D3	8					
U4	1D4	9					
U5	2D1	2					
U6	2D2	3					
U7	2D3	4					
U8	2D4	5					
U9	3D1	12					
U10	3D2	31					
U11	3D3	32					
U12	3D4	33					
U13	4D1	27					
U14	4D2	28					
U15	4D3	29					
U16	4D4	30					
U17	5D1	19					
U18	5D2	20					
U19	5D3	40					
U20	5D4	39					
U21	6D1	14					
U22	6D2	15					
U23	6D3	16					
U24	6D4	17					

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TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U25	1E Input capacitance, enable	23	Open	2	15	pF	19
U26	2E	21					
U27	3E	34					
U28	4E	25					
U29	5E	38					
U30	6E	36					
U31	A Input capacitance, select	35			25		
U32	B Input capacitance, select	13			25		

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NOTES:

1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum.

FIGURE 19. Power supply filter capacitance and input capacitance test circuit for tables XVIII and XIX.

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